LECTURER

- Duc Ta --- dta@sfsu.edu *Office Location:* Thornton Hall 327 *Office Hours:* Before + After class

COURSE

- CSC 340.01 (PiNG) – Programming Methodology; Lecture, 3 units, MoWe, 12:15 PM – 02:10 PM, Thornton Hall 327
- *File Manager (sfsu/student)*: http://csc340.ducta.net
- *Forums (on Canvas)*: https://canvas.sfsu.edu

BULLETIN DESCRIPTION

This course explores advanced data structures and algorithms for manipulating them in C++. Emphasis is placed on design and implementation of those structures and a variety of practical applications. Algorithm coverage will include sorting and searching, and graph algorithms. Students will solve a series of problems to enhance their problem-solving skills.

COURSE PREREQUISITES

- CSC 220, CSC 230, MATH 227, with grades of C or better.

COURSE OBJECTIVES

At the end of this course, students will

- ✓ Be able to write medium-sized C++ programs utilizing STL and an integrated development environment
- ✓ Determine which of the common sorting and searching algorithms to utilize for given problems
- ✓ Be able to apply and implement graph algorithms in practice

COURSE REQUIREMENTS

Students will be dropped from this course in one of the following scenarios:

1. The students **do not have** the course prerequisites.
2. The students failed to show up and participate fully during the **drop period**.

GRADES

Grades will be awarded based on the following breakdown:

- Assignments: 0 pts --- *Due to the budget cuts, the Computer Science department cannot provide grader services.*
    - Students **must** submit assignments at their highest quality to be used as evidence of learning progress.
- Midterm Exam is in-person and in an interview format:
    - **GREEN** Card: **30%** of the course grade **earned**, and the Final Exam is worth **70%**.
    - **YELLOW** Card: **15%** of the course grade **earned**, and the Final Exam is worth **85%**.
    - **RED** Card: **0%** of the course grade **earned**, and the Final Exam is worth **100%**.
    - **No Show**: **30%** of the course grade **lost**, and the Final Exam is worth **70%**.
- Final Exam is a must-take, or the course grade is F (No Show). No make-up exams.

Grades will be determined on the basis of the total possible points of 100 and extra credits received. Letter grade scale:

	A: 90-100%	B: 80-89%	C: 70-79%	D: 60-69%	F: Below 60%
Or	A: 90-100 points	B: 80-89 points	C: 70-79 points	D: 60-69 points	F: Below 60 points
					WU: SFSU-defined

- A and a Letter of Recommendation: 100+% (100+ points), and make the course a better experience for everyone.

ATTENDANCE

- Students are responsible for all material covered in class. It may be a good idea to have a Study Buddy.
- Course announcements and forums are on Canvas. Course materials are organized in the course's File Manager.

LEARNING ACTIVITIES

- Detailed descriptions and instructions to be given in class for each learning activity.
- To make learning activities meaningful and useful, please engage. Different points of view are very welcome.

STUDENT RESOURCES

- Recommended Textbook: Frank M. Carrano, *Data Abstractions and Problem Solving with C++*, 7th edition, Pearson.
- Recommended IDEs: Microsoft Visual Studio Community (default course IDE) and cLion. The latest versions.
- Recommended Services, *please visit the File Manager for more recommended services*:

STUDENTS WITH DISABILITIES

Any students with disabilities or other special needs who need special accommodations in this course are invited to share these concerns or requests with the instructor. The Disability Programs and Resource Center (DPRC) is available to facilitate the reasonable accommodations process. The DPRC is located in the Student Service Building and can be reached by telephone (voice/TTY 415-338-2472) or by email (dprc@sfsu.edu).

TENTATIVE SCHEDULE

DATE	TOPIC	READING and ASSIGNMENT
Week 01 *June 01 + 03*	- Welcome and Hello C++ - **Java to C++**: String Pool, Deep & Shallow Copies, Clone, Memory Management - **Java to C++**: Java Lexicography, Enum, Data Structures, Google Guava - **Java to C++**: Data Design, Class Design Guidelines, Program Flow	ASMT Canvas and ASMT 01 Out Appendices A-B-C-D-E Appendices F-G-H-I-J-K ASMT Canvas and ASMT 01 Due *Independent Reading*
Week 02 *June 08 + 10*	- **Java to C++**: CS Lower-Division-to-Upper-Division Core Concepts - **C++ Basics**: Preprocessor, Compiler, Data Types, Control Structures, String, Array, Multi-dimensional Array, Vector, Enum… - Pointers and Reference Variables - Programming Methodology 1	ASMT 02 Out CH 01 Interlude 02 *Independent Reading* *Independent Reading*
Week 03 *June 15 + 17*	- Functions and Parameter Passing - Dynamic Memory Management 1 - Standard Template Libraries (STL) and Iterators - Classes and Objects: Class, Object, Anonymous Class, Struct, Union	*Independent Reading* CH 06-09 12-14 17-20 Interlude 01+08 ASMT 02 Due and ASMT 03 Out
Week 04 *June 22 + 24*	- Manual Memory Management 2 - Operator Overloading, Copy vs. Move Semantics, the BIG 3 and the BIG 5 - Modern C++: Smart Pointers, Automatic Memory Management 3 - Programming Methodology 2	Interlude 04+06 *Independent Reading* *Independent Reading* *Independent Reading*
Week 05 *June 29 + July 01*	- Array Bag. Template and Generic Programming - **Modern C++**: Smart Pointers and Linked Bag - Recursion	*Independent Reading* CH 02-05 ASMT 03 Due
Week 06 *July 06 + 08*	- **Modern C++**: Inheritance and Polymorphism - **Modern C++**: Multiple Inheritance and Friend Access	*ASMT 04-PREP and ASMT 04 Out* Interlude 05
Week 07 *July 13 + 15*	- Exception Handling - **Modern C++**: Smart Pointers, Cyclic Reference, Slicing Problem - **MIDTERM EXAM**	Interlude 03 ASMT-PREP 04 Due *Independent Reading*
Week 08 *July 20 + 22*	- Sorting Algorithms: The Slow and The Fast - Sorting Algorithms: Complexity Analysis - Programming Methodology 3	CH 10+11+13 15+16 ASMT 04 Due and ASMT 05 Out *Independent Reading*
Week 09 *July 27 + 29*	- Function Pointer, Anonymous Function, Function Object… - **Modern C++**: Lambda Expressions - Final Exam Review	CH 21 *Independent Reading* ASMT 05 Due
Week 10 *Aug 03 + 05*	- **Modern C++**: Lambda Expressions (and Advanced Topics) - Final Exam Review - **FINAL EXAM – WEDNESDAY, August 05, 12:15 PM – 02:10 PM**	*Independent Reading* *Independent Reading* *Independent Reading*

*** The course instructor reserves the right to change any of these policies at any time during the semester. Any changes will be announced in class and posted in the course forums on Canvas.

Course Policy on Student Conduct and Academic Honesty

All students of this course must agree with and strictly follow:

1. The San Francisco State University's **Code of Student Conduct**: https://conduct.sfsu.edu/standards
2. The Computer Science Department's **Student Policies**: https://cs.sfsu.edu/student-policies
3. **The Honor Code** of this course which is detailed below (in this same document).
4. Caring and constructive behavior which help build a safe and healthy atmosphere in the class for everyone will be rewarded. Please ask the instructor for more details.
5. An F grade or a "zero" grade will be assigned for an assignment in which a student violates a rule in **the Honor Code**.
6. An F grade will be assigned for the course to a student who cheats on an exam.
7. An F grade will be assigned for the course to a student who poisons the learning or the testing environment of others, who prejudices the effort of others, or who infringes on the rights of others.

The Honor Code

This Honor Code was adopted from the Computer Science department of Stanford University, https://cs.stanford.edu/degrees/ug/HonorCode.shtml , and was updated.

A. The Honor Code is an undertaking of the students, individually and collectively:

- that they will not give or receive aid in examinations; that they will not give or receive unpermitted aid in classwork, in the preparation of reports, or in any other work that is to be used by the instructor as the basis of grading;
- that they will do their share and take an active part in seeing to it that others as well as themselves uphold the spirit and letter of the Honor Code.

A. Under the Honor Code you are obligated to follow all of the following rules:

Rule 1: You must not look at solutions or program code that are not your own.

It is an act of plagiarism to submit work that is copied or derived from the work of others and submitted as your own. For example, using a solution from the Internet or a solution from another student (past or present) or some other source, in part or in whole, that is not your own work is a violation of the Honor Code. Many Honor Code infractions we see make use of solution code found online. The best way to steer clear of this possibility is not to search for online solutions to the programming assignments. Moreover, looking at someone else's solution code in order to determine how to solve the problem yourself is also an infraction of the Honor Code. In essence, you should not be looking at someone else's code in order to solve the problems in this class. This is not an appropriate way to "check your work," "get a hint," or "see alternative approaches."

Rule 2: You must not share your solution code with other students.

In particular, you should not ask anyone to give you a copy of their code or, conversely, give your code to another student who asks you for it. Similarly, you should not discuss your algorithmic strategies to such an extent that you and your collaborators end up turning in the same code. Moreover, you are expected to take reasonable measures to maintain the privacy of your solutions. For example, you should not leave copies of your work on public computers nor post your solution code on a public website.

Rule 3: You must indicate on your submission any assistance you received.

If you received aid while producing your solution, you should indicate from whom you got help (if that person is not the grader or the instructor of this class) and what help you received. A proper citation should specifically identify the source (e.g., person's name, book title, website URL, etc.) and a clear indication of how this assistance influenced your work (be as specific as possible). For example, you might write "I discussed the approach used for sorting numbers in the sortNumbers method with Mary Smith." If you make use of such assistance without giving proper credit, you may be guilty of plagiarism.

It is also important to make sure that the assistance you receive consists of general advice that does not cross the boundary into having someone else write the actual code or show you their code. It is fine to discuss ideas and strategies, but you should be careful to write your programs on your own, as indicated in Rules 1 and 2.

B. Please be aware: all submissions are subject to automated plagiarism detection.

The course instructor has no desire to create a climate in which students feel as if they are under suspicion. The entire point of the Honor Code is that we all benefit from working in an atmosphere of mutual trust. Students who deliberately take advantage of that trust, however, poison that atmosphere for everyone.

The course grader employs powerful automated plagiarism detection tools that compare assignment submissions with other submissions from the current and previous semesters. The tools also compare submissions against a wide variety of online solutions. These tools are effective at detecting unusual resemblances in programs, which are then further examined by the grader again. The grader then makes the determination whether submissions are deemed to be potential infractions of the Honor Code and should be referred to the Computer Science department.

C. A final note on collaboration.

In computer science courses, it is usually appropriate to ask others--the grader or the instructor, or other students--for hints and debugging help or to talk generally about problem-solving strategies and program structure. In fact, the course instructor strongly encourages you to seek such assistance when you need it. Discuss ideas together, respect the rules together, but do the coding on your own.

STUDENT DISCLOSURES OF SEXUAL VIOLENCE

SF State fosters a campus free of sexual violence including sexual harassment, domestic violence, dating violence, stalking, and/or any form of sex or gender discrimination. If you disclose a personal experience as an SF State student, the course instructor is required to notify the [Dean of Students]. To disclose any such violence confidentially, contact:

- The SAFE Place - (415) 338-2208; http://www.sfsu.edu/~safe_plc/
- Counseling and Psychological Services Center - (415) 338-2208; http://psyservs.sfsu.edu/
- For more information on your rights and available resources: http://titleix.sfsu.edu

COURSE SYLLABUS POLICY, #S23-257

- *Senate Policy*
- *Last review: 03/2023 | Next review: 03/2027*

Check List and Additional Items:

I. Guidelines covering syllabus use in courses
   ✓ Guidelines adopted

II. Basic information for all course syllabi
   A. Provided
   B. Provided
   C. Provided
   D. Provided
   E. Provided
   F. Provided
   G. Provided
   H. Provided
   I. Provided
   J. Provided
   K. Provided
   L. Provided
   M. Provided

   N. Students may not capture audio, photos, or video from class sessions on their own devices without the explicit permission of the instructor and everyone present, unless part of a DPRC-authorized accommodation.

   O. Students may not post any course materials to any third-party sites (such as Chegg) or post any recordings, screenshots, audio or chat transcripts in any setting outside the class, and that violations of this are subject to student disciplinary action.

   P. Provided

III. Additional information for syllabi for courses using asynchronous and synchronous delivery
   ✓ Not applied

IV. Information for course that meet University-wide requirement (i.e., General Education, SF State Studies, and American Institutions requirements)
   ✓ Not applied

V. Statements required by Academic Senate Policy

   A. Disability access
   Students with disabilities who need reasonable accommodations are encouraged to contact the instructor. The [Disability Programs and Resource Center (DPRC)] is available to facilitate the reasonable accommodations process. The [DPRC] is located in the [Student Service Building] and can be reached by telephone [(voice/415-338-2472, video phone/415-335-7210) or by email (dprc@sfsu.edu).]

   B. Student disclosures of sexual violence
   SF State fosters a campus free of sexual violence including sexual harassment, domestic violence, dating violence, stalking, and/or any form of sex or gender discrimination. If you disclose a personal experience as an SF State student, the course instructor is required to notify the [Title IX Coordinator by completing the report form available at http://titleix.sfsu.edu, emailing vpsaem@sfsu.edu or calling 338- 2032]. To disclose any such violence confidentially, contact:

[The SAFE Place - (415) 338-2208; http://www.sfsu.edu/~safe_plc/]

[Counseling and Psychological Services Center - (415) 338- 2208; http://psyservs.sfsu.edu/]

For more information on your rights and available resources: [http://titleix.sfsu.edu]

C. Academic integrity
Academic integrity, the ethical presentation of one's own work in accordance with the rules established for this class, is required. Instances of academic misconduct will be reported to the College in which the course is housed, the Division of Graduate Studies (if a graduate student), and the Office of Student Conduct with the report being kept in those offices until a student earns his/her degree. Any instances of cheating, deceit, fabrication, forgery, plagiarism, unauthorized altering of records or submitting false documents, unauthorized collaboration, unauthorized submission of work previously given credit, or other forms of academic misconduct will be assigned a grade penalty, likely an F or a grade of zero. Failing one or more assignments or examinations for reasons of academic integrity violations may result in a final class grade of F. Students may not withdraw from classes in which they have committed academic misconduct. Consequences for violations of academic integrity may exceed an F on the assignment, examination, or class as determined by the Academic Integrity Review Committee.

Members of our academic community have a responsibility to develop an awareness of academic integrity, to cultivate skills to realize honesty in academic and community work, and to sustain actively academic honor as a core value of our community. Students are expected to engage in behaviors that reflect well upon the university. In addition to attending to one's own actions, the Standards for Student Conduct require that students who witness academic dishonesty notify their faculty/instructor, department chair, or the Office of Student Conduct. Supporting academic integrity enhances the reputation of the University and the value attributed to degrees awarded by the University.